

Refutation of Conjecture 00000000807: the maximal multiplicity of $\lambda_1(\mathbb{T}^2)$ is 6, not 4

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Abstract

We disprove Conjecture 00000000807. For a flat two-torus $\mathbb{R}^2/\Lambda$ the Laplace eigenvalues are $4\pi^2|k|^2$ with k ranging over the dual lattice Λ^* , so the multiplicity of the first non-zero eigenvalue λ_1 equals the number of shortest non-zero vectors of Λ^* . The square torus has multiplicity 4, but the triangular (equilateral) torus has multiplicity 6. The true maximum over all flat tori is 6, attained exactly by lattices similar to the triangular lattice. Consequently both clauses of the conjecture fail.

1 The conjecture

The statement under consideration is:

Conjecture 00000000807. *The maximal multiplicity of $\lambda_1(\mathbb{T}^2)$ is 4 (attained only on the square torus), and at most 3 for general lattices (multiplicity rigidity of lattice spectra, partially known).*

We show that this is false: the maximum is 6, the square torus is not the unique maximiser, and the bound “at most 3 for general lattices” is violated already by the square torus (4) and the triangular torus (6).

2 Spectrum of a flat torus

Let $\Lambda \subset \mathbb{R}^2$ be a full lattice and let

$$\Lambda^* = \{ k \in \mathbb{R}^2 : \langle k, \ell \rangle \in \mathbb{Z} \text{ for all } \ell \in \Lambda \}$$

be its dual lattice. On the flat torus $\mathbb{T}^2 = \mathbb{R}^2/\Lambda$ the Laplace operator has the complete orthonormal system of eigenfunctions

$$\varphi_k(x) = e^{2\pi i \langle k, x \rangle}, \quad k \in \Lambda^*,$$

with

$$-\Delta \varphi_k = 4\pi^2 |k|^2 \varphi_k.$$

Hence the spectrum is $\{ 4\pi^2 |k|^2 : k \in \Lambda^* \}$, and for the first non-zero eigenvalue

$$\text{mult}(\lambda_1(\mathbb{T}^2)) = \#\{ k \in \Lambda^* \setminus \{0\} : |k|^2 = r_1^2 \}, \quad r_1^2 = \min_{0 \neq k \in \Lambda^*} |k|^2. \quad (1)$$

Since k and $-k$ always have the same length, every multiplicity is even.

3 The square torus has multiplicity 4

Take $\Lambda = \mathbb{Z}^2$, so that $\Lambda^* = \mathbb{Z}^2$ as well. The shortest non-zero vectors are

$$\pm e_1 = \pm(1, 0), \quad \pm e_2 = \pm(0, 1),$$

all of squared length 1. Thus

$$\text{mult}(\lambda_1) = 4$$

for the square torus, in agreement with the first numerical claim of the conjecture.

4 The triangular torus has multiplicity 6

Let

$$\Lambda = \mathbb{Z}(1, 0) \oplus \mathbb{Z}\left(\frac{1}{2}, \frac{\sqrt{3}}{2}\right)$$

be the triangular (equilateral) lattice, with basis $b_1 = (1, 0)$, $b_2 = (\frac{1}{2}, \frac{\sqrt{3}}{2})$. Its Gram matrix is

$$G = (\langle b_i, b_j \rangle)_{i,j} = \begin{pmatrix} 1 & 1/2 \\ 1/2 & 1 \end{pmatrix}, \quad \det G = \frac{3}{4}.$$

The dual basis d_1, d_2 , characterised by $\langle d_i, b_j \rangle = \delta_{ij}$, is

$$d_1 = \left(1, -\frac{1}{\sqrt{3}}\right), \quad d_2 = \left(0, \frac{2}{\sqrt{3}}\right),$$

and the Gram matrix of the dual lattice is

$$G^* = G^{-1} = \frac{4}{3} \begin{pmatrix} 1 & -1/2 \\ -1/2 & 1 \end{pmatrix} = \begin{pmatrix} 4/3 & -2/3 \\ -2/3 & 4/3 \end{pmatrix}.$$

Explicitly, for $m, n \in \mathbb{Z}$,

$$|md_1 + nd_2|^2 = \frac{4}{3}(m^2 - mn + n^2).$$

The shortest non-zero vectors $k \in \Lambda^*$ are therefore the six vectors

$$\pm d_1, \quad \pm d_2, \quad \pm(d_1 + d_2),$$

corresponding to $(m, n) = \pm(1, 0), \pm(0, 1), \pm(1, 1)$, and each of them has squared length

$$|d_1|^2 = |d_2|^2 = |d_1 + d_2|^2 = \frac{4}{3}.$$

The next possible length is strictly larger: for instance $|d_1 - d_2|^2 = \frac{4}{3}(1 + 1 + 1) = 4 > \frac{4}{3}$. By (1),

$$\text{mult}(\lambda_1(\mathbb{T}^2)) = 6$$

for the triangular torus. In particular the multiplicity is *not* bounded by 4, and the maximum is *not* attained only on the square torus.

Remark 1. The triangular lattice is realised by the positive definite integral binary quadratic form $Q(a, b) = a^2 + ab + b^2$: the six solutions of $Q = 1$ are exactly $(1, 0), (-1, 0), (0, 1), (0, -1), (1, -1), (-1, 1)$. This is the form whose exact shortest-vector counts are formalised in `lean4/Main.lean`.

5 The true maximum is 6

The following standard argument (the two-dimensional kissing number, in the form used for lattice spectra) shows that no flat torus has multiplicity larger than 6, and identifies the equality case.

Theorem 2. *Let $\Lambda \subset \mathbb{R}^2$ be any lattice and let $r = r_1 = \min_{0 \neq k \in \Lambda^*} |k|$. Then the number of shortest non-zero vectors of Λ^* is at most 6. If it equals 6, then Λ is similar to the triangular lattice.*

Proof. Let $u \neq v$ be two distinct shortest non-zero vectors of Λ^* , so $|u| = |v| = r$. Their difference $u - v$ is a non-zero vector of Λ^* , hence by minimality of r

$$|u - v|^2 \geq r^2.$$

Expanding,

$$|u - v|^2 = |u|^2 + |v|^2 - 2\langle u, v \rangle = 2r^2 - 2\langle u, v \rangle \geq r^2,$$

so $\langle u, v \rangle \leq r^2/2$. Writing $\langle u, v \rangle = r^2 \cos \theta$ with $\theta \in [0, \pi]$ the angle between u and v , this says $\cos \theta \leq \frac{1}{2}$, i.e.

$$\theta \geq 60^\circ. \tag{2}$$

Thus the shortest vectors form N points on the circle of radius r whose pairwise angular separations are at least 60° . Order the points cyclically around the circle and let $g_1, \dots, g_N$ be the N successive gaps, so that $\sum_i g_i = 360^\circ$. If some gap were $< 60^\circ$, then the two points bounding it would have angular separation $g_i < 60^\circ$ (their smaller angle is g_i , since $g_i < 180^\circ$), contradicting (2). Hence every $g_i \geq 60^\circ$ and $60^\circ N \leq \sum_i g_i = 360^\circ$, that is $N \leq 6$.

If $N = 6$, then all N gaps equal 60° , so the six shortest vectors are the vertices of a regular hexagon and the shortest-vector shell of Λ^* is that hexagon. This forces the lattice of shortest vectors to be the hexagonal lattice, i.e. Λ is similar to the triangular lattice. $\square$

Corollary 3. *For a flat two-torus $\mathbb{R}^2/\Lambda$ the multiplicity of λ_1 is one of 2, 4, 6. It is 2 for a generic lattice, 4 for the square type and 6 for the triangular type; the maximum 6 is attained exactly by lattices similar to the triangular lattice.*

Proof. Multiplicities are even, and by Theorem 2 the shortest-vector shell has at most 6 vectors. Since the shortest vectors come in $\pm$ pairs, the possible counts are 2, 4, 6. The square lattice realises 4 and the triangular lattice realises 6; a generic lattice (with no accidental coincidences between lengths) realises 2. $\square$

6 Conclusion

Both clauses of Conjecture 00000000807 are false.

1. The maximal multiplicity of $\lambda_1(\mathbb{T}^2)$ is 6, not 4 (Theorem 2 and Section 4). The value 4 is attained by the square torus, but it is not maximal.
2. The square torus is not the unique lattice attaining the maximum: the triangular torus attains multiplicity $6 > 4$, and in fact the triangular torus is the unique maximiser up to similarity.
3. The clause “at most 3 for general lattices” is false as stated: the square torus has multiplicity 4 and the triangular torus has multiplicity 6.

7 Formal verification

The exact shortest-vector counts are formalised in core Lean 4 (`lean4/Main.lean`, no Mathlib and no `sorry`): `tri_min` proves $a^2 + ab + b^2 \geq 1$ for every non-zero integer pair; `tri_six_shortest` proves that $a^2 + ab + b^2 = 1$ has exactly the six solutions listed above; `square_four_shortest` proves that $a^2 + b^2 = 1$ has exactly four solutions; and `conjecture_00000000807_false` records $4 < 6$ and the existence of a multiplicity strictly larger than 4. The dual-lattice/spectrum correspondence (1) and the 60° /kissing-number argument of Theorem 2 are proved here and reproduced numerically in `reproduce.py`; they are not part of the Lean formalisation.